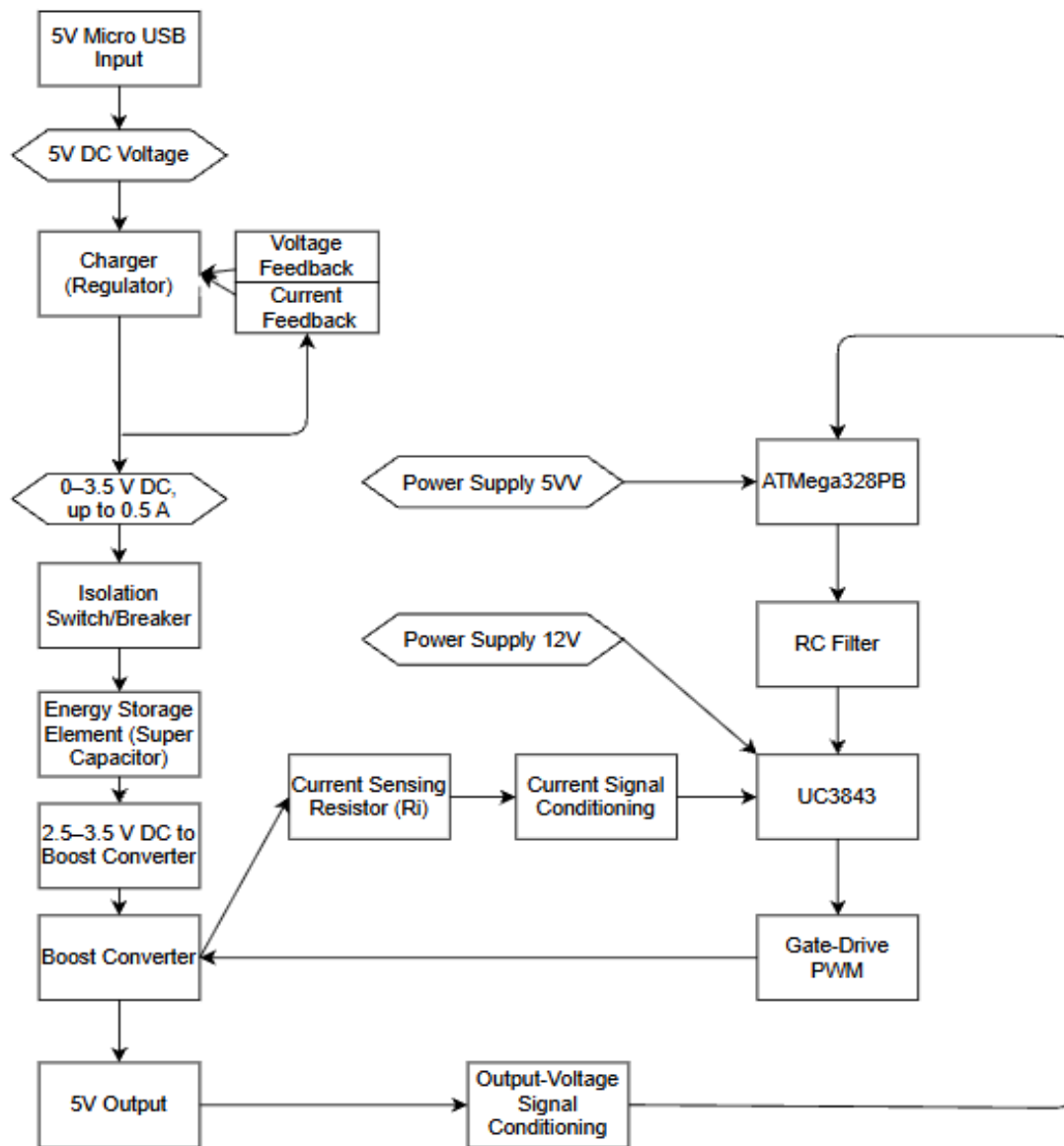
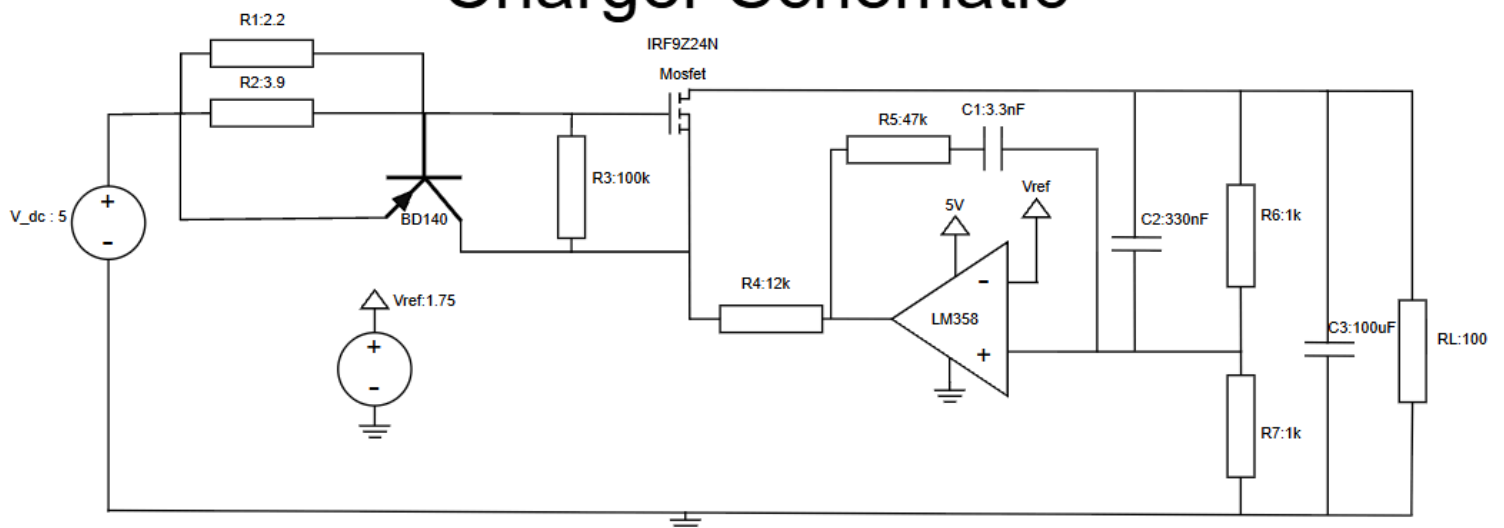


# Project Proposal - Group 9



## Charger Schematic



### Current limiting:

if input current  $> 0.5A$   
 Bit on when  $V_E = V_B + 0.7V(V_{R2})$

$V_{R2} = \text{Input current} \times \text{Resistance}$

$$0.7 = I_{in} \times (2.2k\Omega || 3.9k\Omega)$$

$$I_{in} = 0.7 / 1.4 = 0.5A$$

$R_3$  : Pull up resistor

### Voltage feedback divider:

$$V_{fb} = 1.75V$$

$$V_{fb} = V_{out} \frac{R_5}{R_4 + R_5}$$

$$V_{fb} = V_{out} \frac{1k}{2k}$$

$$V_{fb} = \frac{V_{out}}{2}$$

$$V_{out} = 3.5V$$

.....

### Opamp Loop:

LM358 max output = 3.5V  
 according to the datasheet  
 .....

### Compensation Components:

$$R_9 = 47k, C_2 = 3.3 \times 10^{-9}$$

$$f = \frac{1}{2\pi R_9 C_2} \rightarrow \frac{1}{2\pi \times 47k \times 3.3 \times 10^{-9}}$$

$$f = 1.03kHz$$

### Mosfet:

Mosfet is on when Source  $>$  Gate

Source = 5V (4.3 if current limiting)

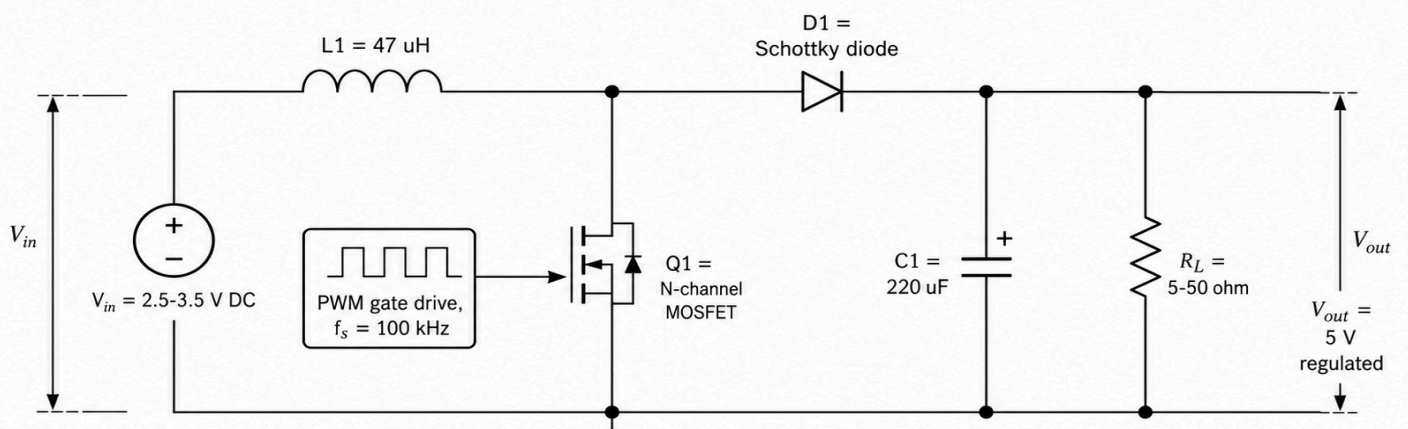
Gate = 3.5V from opamp max output

Drain = 3.5V from Vref voltage divider

Acts as a variable resistor with 1.5V drop

| Parameter              | Value                  |
|------------------------|------------------------|
| Input Voltage          | 5 Vdc                  |
| Output Voltage Range   | 0 Vdc to 3.5Vdc        |
| Voltage Regulation     | 5%                     |
| Output Current Maximum | 0.5 A                  |
| Maximum Output Power   | 1.75 W (at 3.5V, 0.5A) |
| Test Load Resistances  | 50 ohm, 100 ohm        |

## Preliminary Boost Converter Schematic



| Parameter        | Value           |
|------------------|-----------------|
| Output Power     | 0.5 - 5 W       |
| Output Current   | 0.1 - 1.0 A     |
| Duty Cycle Range | D = 0.30 - 0.50 |
| Minimum Load     | 5 $\Omega$      |
| Maximum Load     | 50 $\Omega$     |
| Target Ripple    | < 50 mV         |

The **IRF9Z24N** P-channel MOSFET was selected as the switching device because it provides low conduction losses and simple gate driven requirements in a positive supply system. Compared with a BJT, the MOSFET requires negligible steady-state gate current and is low-resistance minimising voltage-drop, improving efficiency when charging and discharging.

| 30F Super Capacitor                                                                                                        | 150F Super Capacitor                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Smaller (Portable)                                                                                                         | <b>Holds more energy storage (Practical)</b>                                                                                                                                               |
| Lower Cost (\$2.10 v \$4.94 when purchased in bulk(2640 units))                                                            | <b>More representative of a high-capacity power bank</b>                                                                                                                                   |
| Quicker charging allows for more testing / prototyping                                                                     | <b>When the load draws bursts of current, the larger capacitor experiences less instantaneous voltage ripple</b><br><b><math>\Delta V = I\Delta t / C</math></b>                           |
| Safer - should any shorts or faults occur, the energy is less than that of 150F reducing the severity of potential damage. | <b>For the same load current, the capacitor voltage drops more slowly because. A larger capacitance means a smaller voltage drop per second.</b><br><b><math>Q = CV = I\Delta t</math></b> |
|                                                                                                                            | <b>ESR = 100m ohm @1kHz (AC conditions) vs 200m ohm. 150F more efficient as internal heat losses are half that of 30F capacitor.</b><br><b><math>P = I^2 \times ESR</math></b>             |

The **150F supercapacitor** was selected to maximise stored energy and extend the operating time of the power bank. Although it requires a longer charging period, it better demonstrates the capability of the charging circuit and provides a more practical energy reserve.

P-type equivalent of BD179 BJT was the BD180, however it is discontinued, so we have used a **BD140** in the current limiting circuitry as its specs are acceptable ( $1.5A I_c (Max) \gg 0.5A$ ).

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Preliminary Boost Converter Image Generated by <https://chatgpt.com/>

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Task Division:

High level system schematic - Bruce/Kyle

Part search & Feasibility study - Alex/Bruce

Calculations/diagrams - Saketh/Kyle